

Lupus Nephritis — ISN/RPS Classes I–VI (Wolf Den)



1. Anti-dsDNA → complement fixes

WHAT YOU SEE

Wolf holding magnifier + anti-dsDNA chart

WHAT IT ENCODES

Anti-dsDNA titer correlates with renal activity; it fixes complement so active disease drives serum C3/C4 DOWN. Rising dsDNA + falling complement signals a flare. SLE itself is diagnosed by ANA (most sensitive), but dsDNA and anti-Smith are the specific antibodies.

BOARD TRAP

ANA is sensitive, not specific — a positive ANA alone does not predict renal flares; track dsDNA + complement instead.

2. Classes transform → repeat biopsy

WHAT YOU SEE

Door labeled 'classes transform — repeat biopsy'

WHAT IT ENCODES

Lupus nephritis class is not fixed; patients can transition between classes over time (e.g., V to IV). A repeat kidney biopsy guides re-treatment when the clinical picture changes. Biopsy defines class, activity, and chronicity indices that direct immunosuppression.

BOARD TRAP

Don't assume a prior biopsy class still holds at relapse — transformation between classes mandates re-biopsy before escalating therapy.

3. Class I — minimal mesangial

WHAT YOU SEE

Sleeping wolf pup, 'normal LM'

WHAT IT ENCODES

Class I = minimal mesangial: normal light microscopy with mesangial immune deposits only on IF/EM. Clinically silent, normal urinalysis, excellent prognosis. No specific renal therapy beyond treating extrarenal lupus.

BOARD TRAP

Minimal-change disease is a different entity — Class I lupus shows mesangial immune deposits on IF, MCD shows none and only foot-process effacement.

4. Class II — mesangial proliferative

WHAT YOU SEE

Two pups, 'mesangial prolif'

WHAT IT ENCODES

Class II = mesangial proliferative: mesangial hypercellularity + deposits on LM/IF/EM. Mild hematuria/proteinuria, generally good prognosis, usually no aggressive immunosuppression. Watch for transformation to proliferative (III/IV) forms.

BOARD TRAP

Class II does not need cyclophosphamide/MMF induction — reserve aggressive induction for proliferative III/IV.

5. Class III — focal <50%

WHAT YOU SEE

Pup pack, 'focal <50%' subendothelial

WHAT IT ENCODES

Class III = focal proliferative, <50% of glomeruli involved, with subendothelial deposits. Active nephritic picture: hematuria, RBC casts, proteinuria, possible AKI. Induction with high-dose steroids + MMF or cyclophosphamide, then maintenance.

BOARD TRAP

The III/IV cutoff is 50% of glomeruli, not subendothelial vs subepithelial — both III and IV have subendothelial deposits.

6. Class IV — diffuse ≥50% (crescents)

WHAT YOU SEE

Snarling wolf, 'diffuse ≥50% crescents/fibrinoid necrosis'

WHAT IT ENCODES

Class IV = diffuse proliferative, ≥50% of glomeruli, the MOST common and MOST severe class. Subendothelial deposits give 'wire-loop' lesions; crescents and fibrinoid necrosis appear. 'Full house' IF (IgG, IgA, IgM, C3, C1q). Aggressive induction + maintenance immunosuppression.

BOARD TRAP

Wire-loop lesions and full-house IF point to diffuse proliferative (IV), not membranous (V) — V is subepithelial spike/dome deposits.

7. Class V — membranous

WHAT YOU SEE

Wolf with bow tie, 'membranous'

WHAT IT ENCODES

Class V = membranous lupus: subepithelial deposits, diffuse GBM thickening, nephrotic-range proteinuria. Can occur alone or combined with III/IV. Higher risk of renal vein thrombosis (nephrotic state). May be ANA-positive membranous mimicking primary membranous.

BOARD TRAP

Primary membranous is anti-PLA2R positive; lupus membranous (V) is PLA2R-negative with full-house IF and tubuloreticular inclusions on EM.

8. Class VI — sclerotic >90%

WHAT YOU SEE

Petrified wolf tombstone, '>90%'

WHAT IT ENCODES

Class VI = advanced sclerosing: >90% globally sclerosed glomeruli, representing burnt-out, irreversible chronic damage. Immunosuppression is futile; management is ESKD care, RRT, and transplant evaluation.

BOARD TRAP

Don't induce immunosuppression for Class VI — sclerosis is chronic/irreversible; escalating immunosuppression only adds infection risk.

9. APL → microthromboses

WHAT YOU SEE

Blood pool, 'APL → microthromboses, worse despite anticoag'

WHAT IT ENCODES

Coexisting antiphospholipid antibodies cause thrombotic microangiopathy in the glomeruli, worsening renal outcomes independent of the immune-complex class. Anticoagulation is needed for APS thrombosis but the renal TMA can progress despite it.

BOARD TRAP

APS-related renal TMA is a separate lesion from immune-complex nephritis — anticoagulation, not more immunosuppression, addresses the thrombotic component.

10. Hydroxychloroquine — baseline

WHAT YOU SEE

Pill bottle, 'hydroxychloroquine baseline'

WHAT IT ENCODES

Hydroxychloroquine is recommended for all lupus nephritis patients as baseline therapy: reduces flares, improves renal response, and lowers thrombosis risk. Requires baseline + periodic retinal screening for retinopathy.

BOARD TRAP

HCQ is adjunctive baseline therapy, not induction — it does not replace steroids plus MMF/cyclophosphamide for proliferative disease.

11. IV-S MMF / IV-C cyclophosphamide

WHAT YOU SEE

Crescents/maintenance icons, 'MMF / AZA maintenance'

WHAT IT ENCODES

Induction for proliferative LN: high-dose glucocorticoids + either mycophenolate (MMF) or cyclophosphamide; belimumab or voclosporin may be added. Maintenance uses MMF or azathioprine. AZA is preferred when pregnancy is planned.

BOARD TRAP

Cyclophosphamide is gonadotoxic and teratogenic — switch to azathioprine for women planning pregnancy, not MMF (also teratogenic).

12. Avoid prolonged childbearing / pregnancy timing

WHAT YOU SEE

Strongman with axe, 'avoid prolonged childbearing'

WHAT IT ENCODES

Active lupus nephritis worsens pregnancy outcomes and pregnancy can flare LN. Conceive only during ≥6 months of remission, switch MMF/cyclophosphamide to AZA, and continue HCQ. Pre-eclampsia must be distinguished from a renal flare.

BOARD TRAP

Don't continue MMF or cyclophosphamide in pregnancy — both are teratogenic; azathioprine and hydroxychloroquine are the pregnancy-compatible agents.

13. Transplant gate / 20% → ESKD

WHAT YOU SEE

Wall '20% → ESKD', transplant gate

WHAT IT ENCODES

Despite therapy, a meaningful fraction of proliferative LN progresses to ESKD requiring dialysis or transplant. Lupus serologic activity often quiets at ESKD. Transplant outcomes are good; recurrence in the allograft is uncommon.

BOARD TRAP

Reaching ESKD does not require lifelong heavy immunosuppression for the native disease — lupus activity typically abates; manage as transplant/RRT candidate.

14. Renal vein thrombosis (nephrotic)

WHAT YOU SEE

Wolf with red ribbon, 'renal vein thrombosis'

WHAT IT ENCODES

Nephrotic-range proteinuria (especially membranous Class V) causes a hypercoagulable state from urinary antithrombin loss, predisposing to renal vein thrombosis presenting with flank pain and worsening proteinuria.

BOARD TRAP

Sudden flank pain + worsening proteinuria in nephrotic lupus is renal vein thrombosis, not a nephritic flare — image for thrombus before escalating immunosuppression.